Calculus I (MAT 1001), L04

Syllabus

Fall 2026, CUHK-SZ

Instructor:	岑璐宇 Dr Luyu CEN
Office:	志仁楼 Zhiren Building 415
Office hours:	Thursday at 2:00-3:00 pm or by appointment
Email:	cenluyu@cuhk.edu.cn
Lecture Time and Location	Mon & Wed 3:30 am-5:20 pm, Teaching Complex A402

Teaching Assistants:	许游雅 Youya XU	侯骏祥 Junxiang HOU	吕月桐 Yuetong LV
Offices:	TD205	Zhiren 321	RA317C
Office hours:	Thursday 17:00-18:00	Monday 10:00-11:00	Thursday 18:00-19:00
Tutorial time & Location:	Thursday 18:00-21:00 TA301	Thursday 18:30-20:00 TA309	Thursday 19:30-20:00 TC209
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Classes:	T13 and T16	T14	T15

Prerequisites: High School Algebra, Trigonometry, Geometry, in particular, **Chapter 1 of the textbook**

Textbook: *Thomas' Calculus (14th Edition in SI Units)*, George B. Thomas Jr., Maurice D. Weir, Joel R. Hass. Publisher: Pearson/高等教育出版社.

Reference: *Calculus: Early Transcendentals (8 or 9th Edition)*, James Stewart. Publisher: Brooks Cole.

Course Description: This course covers one-variable calculus; it emphasizes intuitive and conceptual understanding of theory of Calculus, computation skills, and nurtures the mentality and the ability to use Calculus to solve problems in other scientific disciplines.

Course Policies:

- The **semester grade** will be given according to performance in
  - homework (10%),
  - weekly quizzes (25%),
  - mid-term (25%) and
  - the final exam (40%).

- **Homework** will be assigned and graded weekly. Each homework assignment covers the sections taught in the lecture class in the same week and must be turned in at the end of the tutorial class of the following week. **Neither electronic copies nor late homework will be accepted. Turn in your homework in your assigned tutorial section. If your tutorial section is T13, take the quiz and hand in your homework in T13, Otherwise, no grade will be given.**
- There will be **no make-up quizzes**—three worst quizzes will be dropped before calculating the semester grade.
- There will be one **midterm exam**, lasting 2 hours. The tentative date of the midterm is **Saturday, Oct 24, from 9:00-11:00 am.**
- Missing the midterm or final exam without **prior notification** to and approval of the instructor will automatically result in the "0" grade for the exam. For the final exam, makeup requests also need subsequent approvals by SSE and Registry.
- Mid-term and final exam papers, as well as homework assignments, will be identical for all lecture sections; quizzes, however, will be created by each course instructor and hence will differ by section; moreover, specific homework problems selected by the instructor and graded by the TAs may also vary by section. To ensure fairness, semester letter grades are assigned using a quota system: at the end of the semester, instructors receive a grade quota for their students based on their rankings in the common mid-term and final exams. Instructors then use the combined, weighted scores (homework, quizzes, mid-term, and final) to finalize letter grades within these quotas.

Course Contents:

Limit of a function and limit laws, one-sided limits, continuity, infinite limits, asymptotes of graphs
Tangents, rate of change and derivative at a point, derivative as a function, higher order derivatives, differentiation rules, derivative of trigonometric functions
Derivative of trigonometric functions(continued), chain rule, implicit differentiation, related rates
Linearization and differentials, extreme values of a function, mean value theorem, monotonicity, first derivative test
Concavity, second derivative test, curve sketching, applied optimization
Newton's method, area approximation by using left, right and mid sums, limits of these sums, definite integral
Fundamental theorem of calculus, antiderivatives and indefinite integrals, substitution method
Substitution method (continued) , area between curves, volumes using cross-sections, arclength, areas of surfaces of revolution
Work and fluid forces, inverse functions and their derivatives, natural log, exponential functions
L' Hôpital' s rule, inverse trigonometric functions, relative rates of growth, integration by parts
Trigonometric integrals, trigonometric substitutions, partial fractions method
Partial fractions method (continued) , numerical integration: by rectangle, trapezoidal and Simpson's rules, improper integrals

First order ODE: solutions, slope fields, Euler's method, linear equations, separable equations
First order ODE: applications--"mixing problems", Malthusian model, Logistic model, autonomous equations, phase-line analysis

Homework Assignments:

2.1: #1, 7, 8, 19, 20
2.2: #1, 4(a)-(e), 9, 10, 18, 19, 23, 28, 32, 40, 44, 45, 54, 57, 65 (a), 75, 77, 80
2.3: skipped
2.4: #1, 5, 6, 17, 22, 30, 34, 36, 48, 49, 50
2.5: #7-10, 26, 27, 32, 41, 45, 49, 59, 61, 64
2.6: #2, 10, 11, 16, 21, 23, 31, 36, 39, 43, 63, 67, 73, 74, 84, 86, 103

3.1: #1, 2, 23, 33, 41, 47
3.2: #1, 20, 27-31, 34, 45,46, 48-50, 59, 60
3.3: #6, 11, 17, 20, 21, 29, 35, 39, 41, 46, 55-58
3.4: #3,7, 15, 21, 23, 25, 26
3.5: #3,12,15,17,26,32, 34, 47, 52,53, 58, 60, 61(a)
3.6: #5, 7, 10, 15, 29, 34, 41, 54, 67, 72, 75, 76, 84, 86,
3.7: #1, 12, 19, 31, 43
3.8: #7, 23, 27, 32, 42
3.9: #1, 6, 17, 18, 29, 37, 42, 51, 56

4.1: #1-6, 20, 36, 42, 45, 59, 60
4.2: #13, 15 a, (iii), 15b, 18, 20, 23, 27, 33, 60, 61, 63,65
4.3: #15-18, 23, 35, 55a, 67
4.4: #1,4,17,23,32,46, 75, 78, 95,97,106,107, 112
4.5: #1, 7, 12, 20a, 22, 28, 42, 52, 53, 55, 58, 71a
4.6: #1, 3, 9, 11, 12, 22
4.7: #1, 7, 12, 23, 32, 34, 52, 59, 63, 75, 85, 99, 103, 107

5.1: #1, 3, 5, 7, 19
5.2: #8,12, 30, 32, 37, 39, 41, 42, 43, 44
5.3: #1, 3, 6, 8, 9, 17,20, 55, 71,73, 76, 77, 78, 81, 83
5.4: #1, 3, 4, 8, 9, 10, 12, 17, 22, 24, 27, 28, 30, 40, 44, 46, 61, 63, 68, 71, 73
5.5: #18, 20, 21, 22, 26, 31, 38, 42, 43, 46, 56, 62
5.6: #8, 11, 12, 21, 23, 25, 59, 63, 71, 72, 78, 82, 83, 84-87
6.1: #1, 10, 12, 14, 21, 25, 29, 31, 39, 43, 45, 53, 61
6.3: #1, 2, 9, 23, 30
6.4: #13, 18, 22, 23, 25, 28
6.5: #3, 9, 12, 16, 25, 26, 31, 38, 46
(6.2 and 6.6 skipped)

7.1: #1-6, 12, 14, 17, 25, 31, 38, 39, 44, 56, 59

7.2: #1, 4, 7, 10, 11, 23, 25, 36, 41, 44, 47, 55, 58, 67, 72, 77, 81, 89
7.3: #1, 8, 16, 18, 26, 28, 29, 33, 42, 52, 60, 63, 66, 69, 78, 79, 86, 89, 92, 95, 100, 103, 115, 122, 131, 133, 141, 146
7.4: delayed
7.5: #16, 20, 21, 26, 31, 38, 40, 41, 45, 51, 53, 59, 60, 61, 66, 67, 68,75, 85, 88
7.8: (taught before 7.6) #1, 9-12, 20, 21(a), 22
7.6: #1, 3, 5, 7, 13, 16, 18, 22, 25, 33, 48, 51, 56, 67, 69, 73, 78, 91, 97, 98, 102, 103, 129,133
(7.7 skipped)

8.1: skipped
8.2: #2,6,12, 17, 23, 28, 29, 31, 32, 37, 43, 45, 50, 51, 53, 65, 67, 75
8.3: #3, 14, 15, 19, 20, 24, 27, 30, 33, 34, 35, 53, 54, 56, 64, 67
8.4: #2, 3, 5, 8, 9, 10, 15, 17, 21, 26, 36, 37, 61, 62
8.5: #6, 7, 13, 16, 20, 23, 27, 31, 34, 35, 41
8.7: #8, 24, 25, 28, 29, 31
8.8: #4, 10, 14, 20, 23, 27, 32, 39, 43, 45, 56, 57, 60, 65, 68, 69, 71, 76, 80

16.1: #1-6, 18 (do 18 after learning 7.4 and 16.2)
7.4: #11, 12, 13, 17, 30, 34, 42
16.2: # 1, 4, 16, 17
16.3: # 13, 15
16.4: #1, 7, 10, 12, 14, 18 (for these problems, no need to study the sign of the second derivative of the solutions)